AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

Test Report No. 2023 – 0031

Agricultural Machinery – Walking-Type Agricultural Tractor HG AIM HG-MBT-YD16

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Test Report Number	2023 – 0031
Date of Issuance	February 28, 2023
Valid Until	February 28, 2028
Agricultural Machinery	Walking-Type Agricultural Tractor
Type	Float-assist tiller, Boat-type, Axle-driven
Brand	HG AIM
Model	HG-MBT-YD16
Test Requested by	H.G. AGRI INDUSTRIAL MACHINERIES Davao City

SUMMARY

Performance Criteria	Requirement as per PNS/BAFS 347:2022	AMTEC Test
Power requirement of the prime mover, kW, maximum	13.5	11.94[a]
Actual Field Capacity, ha/h	ND[b]	0.276
Peak Transmission Efficiency, %, minimum	80	NM[c]
Breakdowns/malfunctions during the five-hour continuous running test.	Shall have none	ND[c]

[a] Data obtained from the prime mover's nameplate used during field performance test.
[b] Not specified by requesting party.
[c] Laboratory performance tests were not conducted.
Note: ND – No Data, NM – Not Measured

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Not provided
c. Operating information	Provided
d. Accessories and attachments	Not provided
e. Maintenance instruction	Not provided
f. Storage	Not provided
g. Handling, reception, transportation, assembly and installation	Provided
h. Specifications	Not provided
i. Dismantling and disposal	Not provided
j. Warranty	Not provided
k. Parts list	Not provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	H.G. AGRI INDUSTRIAL MACHINERIES
4. Prime mover used	11.94 kW YAMADA Gasoline Engine
5. Serial number	No data
6. Number of operators	One
7. Handling and stability of WTAT	The machine is stable when working and turning.
8. Manipulating of the operating levers	The levers are accessible to the operator.
9. Replacing and adjusting of parts	Locally fabricated parts can be replaced. Engine placement can be adjusted.
10. Safety features	Transmission system cover is not provided.
11. Failure or abnormalities that may be observed on the tractor or its component parts.	None
12. The WTAT shall be generally made of steel bars, sheet metals, and other appropriate materials.	The tractor is generally made up of steel materials, GI sheets, and steel pipes.
13. The chain and sprocket used shall conform to PAES 303:2000 (Engineering materials – Roller chains and sprockets for agricultural machines – Specifications and applications) and shall have provisions for refilling of lubricating oil.	The type of chain and sprocket transmission system was not verified. Provisions for refilling of lubricating oil were not verified.
14. The hand grip shall be made of non-slip material and shall have a minimum nominal diameter of 25 mm.	The handle bar is made up of steel pipe with an outside diameter of 26.7 mm.

OBSERVATIONS (Cont'n)

15. Mechanism for handle bar height adjustment with at least three settings shall be provided. The handle bar height shall be 650 mm to 800 mm.	Handle bar height is fixed. The handle bar height was not measured.
16. Mechanism for adjustment of engine speed during operation shall be provided and within reach of the operator.	Engine speed can be manually adjusted on the throttle knob of the engine.
17. Stand assembly for use when the WTAT is not in operation shall be provided.	Stand assembly was not provided.
18. The float shall be made up of steel or other appropriate material. Typical design of floatation structures is shown in Annex A (Common design of floatation structure).	The float is generally made of steel materials and GI sheets. The design of the floatation structure is boat-type
19. There shall be earmuffs or other ear protection device provided for the operator to use when 95 dB(A) is exceeded during operation. Recommendations for use of ear protection device shall likewise be indicated in the operator's manual.	The maximum noise level measured during the field performance test was 95.0 dB(A). Earmuffs or other ear protection device were not provided for the operator.
20. The WTAT shall be free from any manufacturing defects that may be detrimental to its operation.	The WTAT is free from manufacturing defects that may be detrimental to its operation.
21. All metal surfaces shall be painted properly.	All metal surfaces were painted except for the tilling wheel.
22. Parts of the WTAT that are exposed to the operator shall be free from sharp edges and rough surfaces.	Parts of WTAT that are exposed to the operator are free from sharp edges and rough surfaces.
23. Appropriate labels, safety symbols, and warning notices shall be provided in accordance with PNS/BAFS 330:2022 (Technical means for ensuring safety – Guidelines).	Labels, safety symbols and warning notices on the tractor were not provided.
24. All controls shall be in accordance with PNS/BAFS 330:2022 (Technical means for ensuring safety – Guidelines).	Controls with accordance with PNS/BAFS 330:2022 were not verified.
25. All exposed, hot surfaces shall be provided with heat shield or protection to minimize the possibility of inadvertent contact with the operator.	Exhaust outlet of the prime mover was directed away from the operator. However, heat shield or protection was not provided.

OBSERVATIONS (Cont'n)

26. Mechanism for emergency stop of the engine and immediate load disengagement of the transmission system shall be provided.	The power transmission can be disengaged using the main clutch lever.
27. Mechanism for belt transmission adjustment shall be provided.	Belt tension can be adjusted through the main clutch lever or by adjusting the engine base position.
28. Belt cover and mud guard shall be provided. All other moving parts shall have safety guards.	Belt guards were not provided.
29. There shall be a provision for means to minimize vibration.	Provision for means to minimize vibration was not verified.
30. The hexagonal axle shall be in accordance with the specifications of PAES 108:2000 (Agricultural machinery – Hexagonal axle and hub for walking-type agricultural tractor – Specifications).	The axle of the tractor is circular.
31. Other remarks	No markings or labeling were provided on the tractor.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	January 26, 2023
Location of Test	Brgy. Colonia, Valencia City, Bukidnon 7°58'55.21" N, 125°06'44.93" E
Items Tested	Operator's Manual, Depth of Rotary Tilling, Width of Rotary Tilling, Field Capacity, Field Efficiency, Noise Level, Number of Operators, and Fuel Consumption

METHODS OF TEST

The floating tiller was tested based on the PNS/BAFS 348:2022 Agricultural Machinery – Walking-type Agricultural Tractor – Methods of Test.

DESCRIPTION OF THE MACHINE

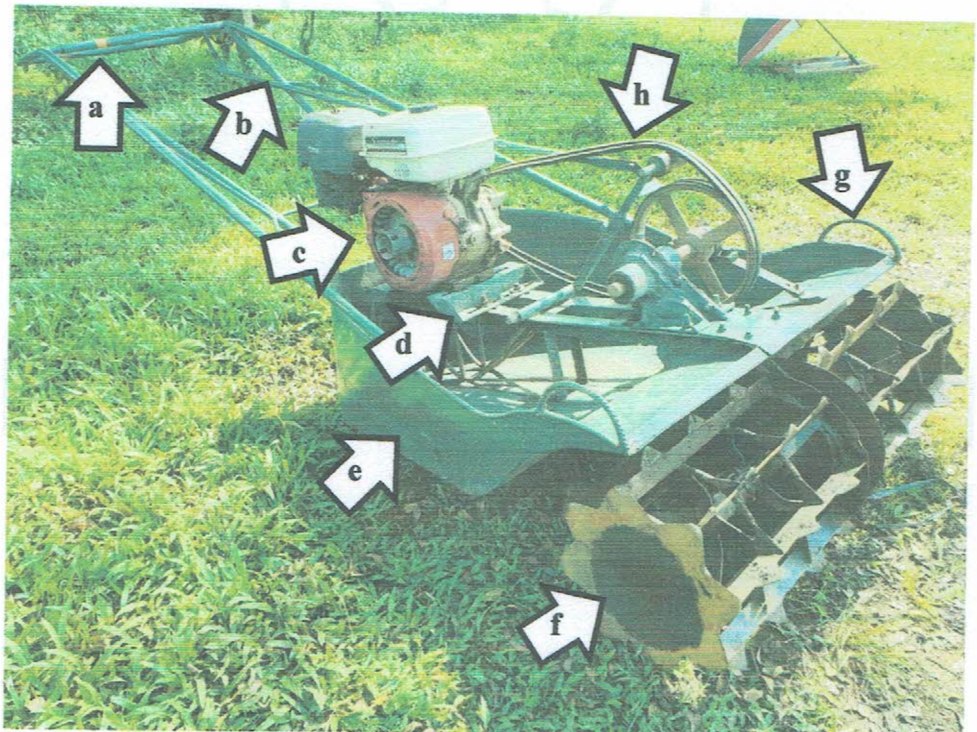

Figure 1. Parts of the HG AIM HG-MBT-YD16 Walking-Type Agricultural Tractor powered by 11.94 kW YAMADA Gasoline Engine include the following: (a) Handle, (b) Main clutch lever, (c) Prime mover, (d) Engine base with mounting holes, (e) Float, (f) Tilling wheel, (g) Lift rings, and (h) V-belt and pulley transmission system.

SPECIFICATIONS

	Item	Manufacturer's Specification[a]	AMTEC Verification
1	Main structure		
1.1	Overall dimensions, mm		
1.1.1	Length	2840	2735
1.1.2	Width	1180	1025
1.1.3	Height	490	875
1.1.4	Ground clearance	ND	NA
1.2	Weight, kg	125	NM
2	Engine		
2.1	Brand	YAMADA	YAMADA
2.2	Model	ND	ND
2.3	Make or manufacturer	ND	ND
2.4	Serial number	GT200701744	GT200701744
2.5	Type	ND	Four-stroke, single inclined cylinder
2.6	Rated power, kW	11.94	Spark ignition
2.7	Rated speed, rpm	ND	NM
2.8	Displacement, cm^3	ND	NM
2.9	Fuel type	Gasoline	Gasoline
2.10	Tank capacity, L	ND	NM
2.11	Fuel consumption, L/h	ND	NM
2.12	Cooling system	Air-cooled	Air-cooled
2.13	Starting system	Rope recoil	Rope-recoil

[a] The data were obtained from the specifications sheet provided by the requesting party.
Note: NA – Not Applicable, ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification[a]	AMTEC Verification
3	Type of clutch		
3.1	Main clutch	Manual	Idler roller
3.2	Steering clutch	ND	NA
3.3	Tilling clutch	Push & pull	NA
4	Power transmission system		
4.1	Engine to input	ND	V-belt and pulley
4.1.1	Engine[b]	101.6 × 2B × ND	101.1 × 2B × 25.0
4.1.2	Input[b]	304.8 × 2B × ND	301.7 × 2B × 29.7
4.1.3	Belt size	B-76	B-76
5	Axle		
5.1	Type	ND	Circular
5.2	Width across the flat, mm	ND	NA
5.3	Diameter, mm	35.0	34.5
5.4	Length, mm	465	635
6	Type of hitch point	ND	NA
7	Rotary tilling wheel		
7.1	Dimension, L × D, mm	475 × 405	472 × 350
7.2	Number of wheels	ND	2
7.3	Number of blades per wheel	ND	8
7.4	Blade dimension, L × W, mm	ND	472 × 35
7.5	Operating width, mm	ND	1025
7.6	Blade teeth		
7.6.1	Shape	Triangle	Trapezoidal
7.6.2	Dimension, L × t, mm	ND	30 × 5.32
7.6.3	Total number of teeth per wheel	28	28 (4-3-4-3-4-3-4-3)

[a] The data were obtained from the specifications sheet provided by the requesting party.
[b] Pulley diameter, mm × number of belts × shaft diameter, mm
Note: NA – Not Applicable, ND – No Data, NM – Not Measured

RESULTS

Item		Data
1	Test conditions	
1.1	Condition of field	
1.1.1	Type	Lowland
1.1.2	Location	Brgy. Colonia, Valencia City, Bukidnon
1.1.3	Soil type	Alimodian clay
1.1.4	Dimensions of field	
1.1.4.1	Length, m	31.5
1.1.4.2	Width, m	16.5
1.1.4.3	Area, m^2	520
1.1.5	Depth of water, mm	33
1.1.6	Number of days soaked	30
1.1.7	Soil resistance, kg/cm^2	NM
1.1.8	Spacing of stubbles	Uneven
1.1.9	Height of stubbles, mm	820
1.1.10	Weed density	NA
2	Field performance	
2.1	Type of field operation	Primary tillage - Rotary tilling
2.2	Field operational pattern	Circuitous pattern rounded corners
2.3	Travelling speed, kph	3.00
2.4	Average depth of rotary tilling, mm	98
2.5	Average width of rotary tilling, mm	894
2.6	Operating width, mm	1025
2.7	Actual field capacity	
2.7.1	ha/h	0.276
2.7.2	h/ha	3.63
2.8	Theoretical field capacity	
2.8.1	ha/h	0.308
2.8.2	h/ha	3.25
2.9	Field efficiency, %	89.56
2.10	Fuel consumption	
2.10.1	L/h	1.61
2.10.2	L/ha	6.20
2.11	Noise at operator's ear level, dB(A)	95.0
2.12	Overlapped, %	24.66

Note: NA – Not Applicable, NM – Not Measured

OBSERVATIONS

Figure 2. Quality of work.

Figure 3. Markings on the prime mover.

Tested by:

GLORY YSTER M. REGINIO
Test Engineer

Verified by:

RAY D. ADIPIO
Test Engineer

Approved for Release:

ARTHUR L. FAJARDO
AMTEC Director

MAR 1 0 2023
Date

REMINDER

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